Pyth on语言程序设计_7c复习

1 P yth o n 12

1.1 P yth o n 12

1.1.1   12   

1.1.3 P yth o n15





!

"#$%

&'

()*+

(),

-./%0

1234

567

8

9: ;

0<=>

?9@AB

1.2 P yth o n CD

A n aco n d a3EFGH9p yth o n ICD

P yC h arm EJK ICD

LM6A n aco n d a3 + P ych arm

NO./4PQO&

p ipPQEp ip in stall+4R

co n d aPQEco n d a in stall+4R

PQp ym ysq lEp ip in stall p ym ysq l

S conda install pym ysql

TUM6co n d a PQ

P yth o n OV WXVYZ[V

1.3 \P yth o n]EH e llo W o rld 24

1.3.1 ^Z _ 25

M6`ab \cd9 ;ef59gh

;^ij [ ], { } S ( ) klmnopM659qrs

P yth o n -tM6ul(')wxul(")wNul('''vSv""") yzh{s|

1.3.2 9} 25 #

1.3.3 59} 26 (N\ul) ’’’ S bN\xulc ”””

1.3.4 ~ 26

P yth o n]_ =>y_

?swwlTUM6e n_ 
wnTU\9t-%

1.3.5 ¡Y ¢ 27

input(‘…… ¡£h¤¥……’) ¦6r§r§¨© ¡ª«¬-{s3

®¯op ¡°S±²³opM6intSflo at>9´

p rin t²t{s|µV)¶<$ ¢·¯¸¹

VE print({s|1,{s|2,{s|3,…,se p = 'eºs')

1.4 ID LECD 29 »¼½¾VESh e llYEd ito r

1.6 L¿ÀÁ 32

2 Pyth onÂ & 34

2.1 ÃÄ 34

2.1.1 \ÃÄ 34

ÃÄR = Å

ÆP yth o n^²nopÇÈÉÊÃÄRËÌ3 ²ÍrÎÅÏ-ÐTÑ»3 *+
ÃÄ

ÆP yth o n^²nÒÃÄÅ-tÃÓ²ÃÄ3 Ô-tÕÖ×Ã²P yth o nØ
ÙÚªÎÅS?yÛÜÝÞÃÄ3 

ÃÄßR³E

à{w{áwâãäåK

n{ æ

nM6¤çè{

éêeë±

ÃÄßRÀìE

íRîï

ëðñEò\óô{áõë²ö®EM yN am e

±ðñE÷\óø²ò\óô{áõë²ö®Em yN am e

âãäEö®Em y_n am e

2.1.2 Ö5\ÃÄ 35 >>> a, b, c = 1, 2, "Teresa"

Code _ùÊE\° *+1Y2eúûÃÄ a Y b²{s|*+ "Teresa" e
úûÃÄ c

2.2 ª3 36

{3

{ª3 6üýþÅ

ÿĀn-×Ãª3 ²āïĂă×Ã{ª3 Ùeú\Ą*+

intbąsl° ²ĆćĈĉf° S°²ĊSċ°²nČ±c

° nµV´č

b in ()EĎ><´čfď><² ¢µV{s|

o ct()EĎ><´čfĐ><² ¢µV{s|

h e x()EĎ><´čfĎđ><² ¢µV{s|

int() Er§\sĒ° {s|²«d{s|´čf°

° zhµV

ď><Et“0 B”S“0 b” æbF®0b101c

Đ><Et “0o”S“0 O ” æbF®0o510c

Ď><EF®10w29w40

Ďđ>< Et“0 x”S“0 X” æbF®0xA 7Ac

flo atē

P yth o n^ē ĔtĎ><zh²à°Y±ĕeb-t0cåK

*üĖëSėĖ±ē ²-tM6Ę ę&zh

P yth o n^ē xĚA²ò\ē ªě8\{ĜbÏ64ĝc

co m p le xĞ

Ğà“ğĕ”Y“Ġĕ”ĕeåKġ
-t6a + b j,Sėco m p le x(a,b)zh

ğĕeYĠĕeõē ġ

Ġĕe¿(ĢģąjSJ

x = 3 + 4 j #M6jSJzhĞĠĕ
y = complex(5,6) #5+6j
x + y #·¯(8 + 1 0 j)
x.im ag #Ġĕ4.0
x.re al #ğĕ3.0

boolĤĥ3

P yth o n^Ĥĥ3 ĦąTru eYFalse\ħÅġ

Tru e*Ĩ°1²False*Ĩ°0

ĩüFalseÅE

N oneġ

Falseġ

Īīf0{3 ²®0w0.0w0jġ

Īī]Ĭ²®''''w()w[]ġ

{ĭ²®{}ġ

6Į3ğö²®3^į__b o o l__()Sė__le n __()O&²«İ¦O&¬-0
SėĤĥÅFalse

{3 ´č

åĒ3

{s|E 'hello Python'

{s|S|(Strin g)à{w{áwâãäåK|{sÿĀn-×Ã
ª3 

ĔıĲzhfE

> > > s =‘A B C D E……’
> > > s = “A B C D E……”

  ^zhZª3 

{|OV:

9{s|E9{s|ijÆ*ulS*xul^

59{s|E59{s|t*NulSNxulĳfĴĵyzh 

ÐT{s|

ĦpfÃÄeú\ÅÏ-

ö®E
var1 = 'Hello World!'

var2 = "Python Programming "

P yth o n{|Ĭzą 2»ħÅĶ]:

ķĸĹĺâĻļuĽľ0 Ŀ²ŀëŁ({s|łA-1)

ķĺĹĸâĻļuĽľ-1 Ŀ²ŀëŁ{s| æ

Ńń{s|

M6ÃÄv[ĿļuE·ŅļuEņł]²m-tŇħňĨ{s|²Ì^âĻķ
0 Ŀŉ²-tĊSċ²âĻ-tfzhħĹæSŊ

nij·ŅĝçâĻ*Ĩª²Ċċŋ-

ņłŌħńÁ²Ċċŋ-²Ľľņłf1

´{s

VÓ ¢

>>>pi = 3.141592653
> > > p rin t('% 1 0.3 f' % p i) #{ōŎ10²ĚA3
3.142
> > > p rin t('% -1 0.3 f' % p i) #ĸ*ŏ
3.142

p rin t('ŐR{%s²Ő lE%s²ŐőŒ%sœ' % (nam e, sno, age))

p rin t(f'ŐR{{n am e }²Ő lE{sn o }²ŐőŒ{age }œ')

Ìÿ¤ç´č str1 = '10'
print(type(eval(str1))) #<class 'int'>

ĬzE[3,‘s’,“A lp 3”]

ŔåE(3, ‘s’, “A lp 3”)

{ĭE{“^ŕ”:“Ŗŗ”,“Řŕ”:“řŚ”}

JĒE{'t', 'n ', 'h ', 'o ', 'p y'}

2.2.1 Number 36

2.2.2 String 37

2.2.3 List 39

2.2.4 Tuple 41

2.2.5 Set 42

2.2.6 Dictionary 43

 m ath

1. ć

2. Åzh

3. śY*

4. NŜ

5. ŝĩŞ

2.3?s 44

2.3.1 ěĝ?s 44

2.3.2 ş?s 45

2.3.3 ÎÅ?s 46

2.3.4 FG?s 47

hö

b1c{s|Y{nFG²Šą-F%nª3 >9FGÔÙ¢
š
> > > ‘H e llo’ > 3 # &šŢ
> > > [1,2]> 5 # &šŢ
b2c÷įÅ-tFG²{s|wĬzwJĒÔ-t>9FG?
> > > ‘ap p le ' > ‘an t' #FG{s|ë±
True
> > > [1, 2, 4] < [1, 2, 3] #FGĬzë±
False
> > > {1, 2, 3 } = = {3, 2, 1 } #ţŤ\JĒťňĩ
True

> > > {1, 2, 3 } < {1, 2, 3, 4 } #ţŤťŦJ
True

2.3.5 ?s 48

hö

?s?Ĥĥ ª²¬-·¯ÔĤĥ ª
n, a = 80, 100
print(n >= 0 and n <= a)
print(0 <= n <= a)

2.3.6 ĝ?s 48

ŧS?%

ĪïYÛŨŧS·¯f0ġ0YĪïŧS·¯ũÌŨ
M6ĝ?³®â:
a ^= b #a = a ^ b
b ^= a # b = b ^ a ^ b Ūfb ^ b = 0 ňūüb = a
a ^= b # a = a ^ b ^ a ňūü a = b
print(a, b)

2.3.7 ŨŬ?s 50

}ï

ū*+ÅňÖ²ĦąÌfÅ S{s| ²a is bŭfTru e²ūaYb
tu p le²list²d ictSse t Ö²a is bfFalse 

2.3.8 KŮ?s 51

KŮ?s6üůÞ\-Ű*+b]ĬwJĒw{ĭc^ťijű\ŔŲ

höE
str1, str2 = 'Py', 'Python'
print(str1 in str2)
x, y = 15, ['abc', 15, True]
print(x in y)
x, y = 2, (1, 2, 4, 6)
print(x in y)
x, y = 'one', {'one' : 1, 'two' : 2, 'three' : 3}
print(x in y)
print(1 in y)

¢E
True
True
True
True
False

2.3.9]Ĭ?s 52

str1, str2 = 'I lo ve P yth o nų', 'I love progam m ing!'
print(str1 + str2)
list1, list2 = [12, False], ['abc', 15, True]
print(list1 + list2)
t1, t2 = (1, 2, 3), ('one', 'two', 'three')
print(t1 + t2)
str1_3 = str1 * 3
print(str1_3)
x_3 = list1 * 3
print(x_3)

¢E
I lo ve P yth o nųI lo ve p ro gam m in g!
[12, False, 'abc', 15, True]
(1, 2, 3, 'one', 'two', 'three')
I lo ve P yth o nųI lo ve P yth o nųI lo ve P yth o nų
[12, False, 12, False, 12, False]

2.3.1 0?sÈŴ 52 ŵ->ŝ

2.4:[; 53

2.4.1 ifwe lifwe lse 55

eŶ·ŷEif ;

Ÿif ;^ůÞ:[KŹ²ź9if ;¿_ōġ
ŸůÞ:[nKŹ²³Ż,if ;¿_ō
eŶ·ŷ^_ōĦą“ź9”“Ż,”»ıĲ

ďeŶ·ŷEif-else ;

Ÿif ;^ůÞ:[KŹ²ź9_ō1
ŸůÞ:[nKŹ²³ź9_ō2

5eŶ·ŷEif-elif-else ;

if ůÞ:[1:
_ō1
e lif ůÞ:[2:
_ō2
……

e lif ůÞ:[n:
_ōn
else:
_ōn+1

eŶżŽžeŶ·ŷ¤ĕijeŶ·ŷ
if ůÞ:[1E
_ō1
if ůÞ:[2:
_ō2
_ō3

2.4.2 pass 56

2.5 ſC ; 57

2.5.1 fo rſC 58

ƀƁſCEfo rſC

ƂĻ-t{s|wZ[wrange()SåĒª3 ĩġ
ſCÃÄ6üƃýƄſC^ŃńĹƀƁ·ŷ^ŔŲġ
fo rſCſCƄħƅüƀƁƂĻŔŲ\

höEM6fo rſCƀƁ{s|
str = in p u t('Ɔ ¡\{s|E')
for c in str:
print(c, end=' ')#tº ²Ľľč9

Ɔ ¡\{s|Esad saf
s a d s a f

fo rſC-tƀƁrange()ÐT°Ĭz

range([start,]stop[,step])
jE
startEzhĬzŉĿĝç²¦-tƇƈ²ƉÖĬzĽľķ0 Ŀ
sto pEzhĬz·Ņĝç²Ɗniksto pö®range(5)wrange(0,5)zhĬz
[0,1,2,3,4]
ste pEzhĬz^ŔŲƋ²¦-tƇƈ²ƉÖĬzņłĽľf1

M6fo rſCƀƁ{s|
str = in p u t('Ɔ ¡\{s|E')
for i in range(len(str)):
print(str[i], end=' ')

Ɔ ¡\{s|Esd aasd
s d a a s d

M6fo rſCğęnƌƍ
n = in t(in p u t("Ɔ ¡\{:"))
fact = 1

for i in range(1,n+1):
fact*=i
print('n!= %d' %fact)

Ɔ ¡\{:4
n!= 24

2.5.2 w h ileſC 59

:[ſCEw h ileſC

ŸſC:[fTru e²³ſCź9w h ileſC^_ōġ
ŸſC:[fFalse²ƎƏw h ileſC
Ÿw h ileſC:[ƐTru e²ā»ıĲƑƒƓſC 

P yth o nw h ileſCYfo rſC¿(-tąe lseeŶ
höEM6w h ileſCğęnƌƍ
n = in t(in p u t('Ɔ ¡\°E'))
fact = 1
i = 1
while i <= n:
fact *= i
i += 1
else:
print('ęƔK²ſCĊć·Ņ')
print('n!= %d' %fact)

Ɔ ¡\°E3
ęƔK²ſCĊć·Ņ
n!= 6

ÛàƕÁ

2.5.3 ļu 60

2.5.4 break 61

2.5.5 continue 62

2.5.6 else 62

2.6 ±· 63

2.7 L¿ÀÁ 63

L¿ğƖ

¾Ɨ\Ƙ6Įƙƚ
demo04_login.py
pƛE ƜlEad m in Ɲ_E123
ƞƟoƛE
1. ƙƚÖû3ƄƠÙ
2. ®¯K²ghơƢxxx
3. ®¯ƙƚƣƤ²ghƚ¡šŢƥũąxƄƠÙ®¯3ƄƠÙM6ƔƦ²³gh
ƙƚƧƨ²ƆÊƩƪƙƚ
èƫşeƬE
1. ƙƚop6ĮRYƝ_²Ôm\{s|
2. 6ĮRYƝ_Ĩ¦M6èƭ ¡²Ʈħ\{s|
3. ƯưƱŭƙƚKƲop}ƳÖƴƵM66ĮRYƝ_
4. Ɩƶ ¡6ĮRYƝ_Y}ƳÖťƷ
5. 3ƄƠÙ²M6fo r
6. ®¯ƙƚKopŻ¢ſCghơƢxxx®¯ƣƤ²opifůÞťƠÙM
6ƔƦ

register_username='admin'
register_password='123456'
for i in range(3):
user_nam e = input("Ɔ ¡6ĮRE")
user_passw ord = input("Ɔ ¡Ɲ_E")
if user_name==register_username and user_password==register_password:
print("ƙƚK²ơƢ% s"% u se r_n am e)
break
else:
if i==2:
print('ƙƚƧƨ²ƆÊƩƪƙƚ!')
else:
print('ƙƚƣƤ²Ƹũą%dƄƠÙ!' % (3 -i-1))
else:
print('ƙƚƣƤų')

¾ƗƹƺƻƼĬ ¢
demo05_fibonacci.py
ƞƟoƛE
1. 6Į ¡žĬŁ
2. Ċƽ ¢·¯
èƫşeƬE
M6inputrƾ6Į ¡
M6w h ileſCğ

n = in t(in p u t("Ɔ ¡žĬŁE"))
a=0
b=1
i=0
while i<n:
print(b,end=',')
a,b=b, b+a
i=i+1
#Ɔ ¡žĬŁE6
#1,1,2,3,5,8,

ƿ9ǀǁëǂ{´č
ƞƟoƛE
ƿ9ǃŦŶǄƞƟÆǀǁĕeopM6ëǂ{²ŪƉopd6Įƚ¡{
¸¹´Ãfǂ{
ƂǅĦoƔK1~5ĝ°´čÏ-
höE
¡ǀǁ:> 3 2 5 4 2
ǂ{´č:> ǆǇǈǉǊǋǌǍǈ ǎ °
èƫşeƬE
M6ForſCƔK{òĝǏ
ǐ6ĬzâĻğ*ĝ´č
Ǒǒ
]-tǏef3\CĜğE
CĜ1Eę¢6Į ¡ǀǁĝġ
CĜ2Eǐ6ǓîĝƔKòĝǏġ
CĜ3EĆ,ĬzâĻ*ĝğŀƎ ¢
IƫǔE
opÐT\Ĭz²f¿Ǖ*ĝ´čƒǖǗE
ǂ{ĬzE['ǘ', 'Ǚ', 'ǈ', 'ǆ', 'ǌ', 'Ǌ', 'ǚ', 'Ǜ', 'ǜ', 'ǝ', 'Ǎ']
ĝĬzE['ǎ','Ǎ', 'ǋ', 'ǉ', 'Ǉ']

n u m ch ar= ['ǘ', 'Ǚ', 'ǈ', 'ǆ', 'ǌ', 'Ǌ', 'ǚ', 'Ǜ', 'ǜ', 'ǝ']
p r= ['ǎ','Ǎ', 'ǋ', 'ǉ', 'Ǉ']
n u m b e r= in p u t("Ɔ ¡\ëüǘnƧ,Ǟĝ{E")
lengh=len(number)
for i in str(number):
lengh-=1
print('%s%s'%(numchar[int(i)],pr[lengh]),end='')
#Ɔ ¡\ëüǘnƧ,Ǟĝ{E312
#ǆǋǙǍǈǎ

]² ¢ǟǟƍ&z²®âǠƵh

for i in range(1,10):
for j in range(1,i+1):
print(j,"*",i,"=",i*j,end=' ')
print('\n')

]²ƅ“ǡǢǡǣ”ńÁ
 ĦǤǣÅǞǢ²ĦáǣÅNǢ²NĦ±ǣÅǢ

for i in range(1,101):
for j in range(1,101):
if i/3 +j*3+(100-i-j)*5==100 and 100-i-j>0:
print('±ǣE',i,'áǣE',j,'ǤǣE',100-i-j)
# ±ǣE 78 áǣE 18 ǤǣE 4
# ±ǣE 81 áǣE 11 ǤǣE 8
# ±ǣE 84 áǣE 4 ǤǣE 12

āą1YŨ°÷°ƑŲġ®\Ųķĸ)ĺYķĺ)ĸň²
³¦Ųf-ZŲ ƛ¢2 -1 0 0 0¤Ƶą-ZŲ

for n in range(2,1001):
m=int(n**0.5)
i=2
while i<=m:
if n%i==0:
break
i+=1
s1=str(n)
s2=s1[::-1]
if i>m and s2==s1:

print(n,end=' ')
#2 3 5 7 11 101 131 151 181 191 313 353 373 383 727 757 787 797 919 929

3  68

3.1 ǥ6 68

3.2 Ĭz¬-Å 69

3.2.1 µ 70

µĶ]Eĝç>ČǦ>Ľľ>ČǦ
def func(name,*args,age=10,**kwargs)

3.2.2 ğ 70

ĝçÆǅ²è{Æ¿
func(25, c=24)

3.2.3 Ľľ 71

3.2.4 è{ 72

3.2.5 nł 73 µ^\ǦŔå²\Ǧ¢y{ĭ

3.2.6 ǏeĬz 75

®¯\ƵopýþÆĬzwŔåS{ĭ^²m-tÍrķĬzwŔå
S{ĭ^Ǐe¢Ƶop²Ì^ĬzwŔåǏe¢y·¯ĳfĝç
²ǧ{ĭǏe¢y·¯ĳfè{

3.2.7 ¬-Å 76 return zh¬-²¿Ǖ_õnÙĈź9

3.3 ¾ 77

3.3.1 import 78

im portè{Ǩ¡¾ 
Im p o rt m ath as m #Ǩ¡m ath¾ «ŉǩRm
im port sys, tim e #ƄǨ¡sys,tim e\¾

3.3.2 from import 81

Ɖø²ŸĦǪǨ¡¾ ^ű*+²³-tM6fro m Ǩ¡¾ ^ž*+²Ì
&V®âE
fro m ¾ R im port Ǩ¡*+R #Ǩ¡¾ ^ű*+
Ô-tfǨ¡*+ŉǩR² &fE
fro m ¾ R im port Ǩ¡*+RasǩR #ûǨ¡*+žǩR
®¯ǪǨ¡¾ ^Ƶą*+ ²-tfE
fro m ¾ R im port *
}ïEM6fro m Ǩ¡*+-tÍrM6²nopM6¾ Rĳfƨs

3.3.3 i 82

\ià5\¾ åK²Ïą5\.p yZ[²ƻưā\Ƶǫ“i”mŐĀǬǭ
\ƂƚǮįÆopƅ®īu6ű\Ƃƚ^¾ ńÁƅO&mÆ¦
Ƃƚ^ǯ__in it__.p yZ[ __init__.py\Z[²dǯÆű\Ƃƚ^²
m-td¦Ƃƚ^Ìÿ.p yZ[ĳf¾ Ĉu6
£hE__in it__.p y-t\Z[²Ô-tijiĿÓ_Sėǰç__all__
Ĭz

3.3.4 ǱŦǲǳ 83 ǱŦǲǳžÆ?9ÖÜǴǵčǓą_²ǧnopǶ×ǷĿ_

3.3.5 NO¾ ƮħPQ 84

random bc¬-\0Ĺ1ÕƠēn (0 < = n < 1)²¦ÆğǸńÁ
^ąăǹćǺǻĨ6

2wtim e¾
tim e¾ 6üƮħ«ǼǽÖ²p yth o n^ą»ÖzhOV²
1cÖǾ
ÖǾžķǿȀȁÖ1970Œ01Ȃ01ȃ00Ö00e00ȄŉȅÆƐȄ
tim e¾ ^tim e ()-tƮħūǅÖǾȆâö4.
tim e¾ ^tim e ()6&
import time
P rin tbtim e.tim e ()c

Æhö^²ôÈǨ¡tim e¾ ²Ć,tim e¾ ^tim e ()ƮħūǅÖÖ
Ǿ²?9·¯fEķ·¯^nÍrȇ¢ƵzhÖ²-td·¯´čfÖ
Ŕå²ƪ>9V ¢

2cÖŔå
ÖŔåstru ct_tim eij9\ŔŲ² tm _year,tm _m on,tm _dayĩ²ò\{ōzh
nj²tm _ye arzhŒŬ²ħÅŁE0 0 0 0 -9 9 9 9°²tm _m o nzhȂ
Ŭ²ħÅŁ1 -1 2°²tm _d ayzhȃȈ²ħÅŁ1 -3 1°²Ìÿ{ō
jËħÅŁ®z1ƵhÆtim e¾ ^²localtim e()-td\ÖǾ´č
f\ūǅÖêÖŔå
ÖVÓsl

3wsys¾
ā\ȉp yth o nØƝȊĻǖ4
import sys
print(sys.__doc__)
ghįsysZȋ²;Ȍkį¾ 
1csys.argv
sys.argv ÃÄ²ȍ6y)p yth o nØȎȏ²ƵtRȐ“ßȑ9”
import sys
p rin t(“th e file n am e:”,sys.argv[0])
p rin t(“th e n u m b e r o f argu m e n t”,le n (sys.argv)
p rin t(“th e argu m e n t is:”,str(sys.argv)
dȒ(_ƃý²Z[R1 1 2 1.p yȓ¿Æßȑ¼½?9²?9·¯
the file name:1121.py
the number of argument 1
th e argu m e n t is: [‘1 1 2 1.p y’]
2) sys.exit()
ā\O&ïǑȔ¢ūǅ]

3.4 ÃÄĳ6ȕ 84

3.4.1 ȖĕÃÄ 85

3.4.2 ȗȖÃÄ 85

3.4.3 glo b alè{ 86 Ș¦ȗȖșȚu,y²ĔnÕïǶ×

3.4.4 n o n lo calè{ 87 ÆȖĕ²țȜȝŀȞøȟÃÄșȚȠ¤¥ȡȢǶ×

3.5 ŝŴĨ6 88

3.5.1 ȏȣ 88

3.5.2 ŝƌ 89 ŝƌžȘĳf»

3.5.3 lam b d a 89

lam bdaÔĉfȤR²»nM6defµV²Ìĳ6ȥ@
\Ȧ

lam bdaĦ\zȧV²Ƶtlam bdaĆćĦğFG


3.5.4 Ȩi 90

®¯¤ȟM6įøȟ^ȖĕÃÄ²«İøȟ¬-Å¤ȟ
u6²mŷKįȨi

Æøȟ^Ɗà¤ȟM6ÃÄĈĉfÛàÃÄ

ĔıĲâ²®¯\·Ņ²ƻư¦^ȖĕÃÄmõÙǯ

3.5.5 QȩØ 91

ǐ6QȩØ²-tÆnǶ×ǓąıĲâ)Ǔą^}¡_²MÌȪǗĄ


\QȩØ-tf5\}¡_²\Ô-t}¡5\QȩØ_

ǐ6QȩØ-tdȃȫǼǽwź9ÖęĩGfĆ6_}¡Ĺn
^²ķǧMȇ_Ȭ'

3.6 ±· 94

3.7 L¿ÀÁ 94

L¿ğƖ

'''

ƞƟoƛE
ŔďƄO^ ^î²Ĕyȭ&ąǤV&wŪVe&
ĩ
-tÚªÛȮǽ²ōƛŔďƄO]
èƫşeƬE
ŀǑǒ6ǤV&ƛ²āȯȰ&³
'''

import math
a= flo at(in p u t("Ɔ ¡aÅE"))
b = flo at(in p u t("Ɔ ¡bÅE"))
c= flo at(in p u t("Ɔ ¡cÅE"))

x1=((-b+math.sqrt(b*b-4*a*c))/(2*a))
x2=((-b-math.sqrt(b*b-4*a*c))/(2*a))
p rin t('O:% d x^2 + % d b + % dÚx1=% d,Úx2 = % d '% (a,b,c,x1,x2))

#?9·¯E
Ɔ ¡aÅE1
Ɔ ¡bÅE-3
Ɔ ¡cÅE2
O:1 x^2 + -3 b + 2Úx1=2,Úx2=1

# 1. ²ȱȲƮħȎ¡ĬzSėŔȳ*+Ƶąȴfļu*ĨŔŲ
def odd1(s1):
li=[]
for i in range(len(s1)):
if i%2==1:
li.append(s1[i])
return li
li1=odd1([1,2,3,4,5])
tup=odd1((1,2,3,4,5))
p rin t('ūȎ¡ĬzÖ ¢fE% s'% li1)
p rin t('ūȎ¡ŔåÖ ¢fE% s'% tu p)

# 2. ²ůÞ6ĮȎ¡*+b{s|wŔȳwĬzcłAťëü6
def len_if(args):
if(len(args)>6):
print("Ȏ¡*+łAëü6 ")
else:print("Ȏ¡*+łAnëü6 ")
a= in p u t("Ɔ ¡půÞłA(ťëü6c*+E")
len_if(a)

3. ²ȱȲȎ¡ĬzłA²®¯ëü2²dĬzǅȵ¤¥¬-ûǥ6
ė
def len1(*args):
for i in args:
if(len(i)>2):
return (i[:2])
else:
return (i)
print(len1([1,2,3]))

4. ²ęȎ¡{s|^²{w{áwtËÌÿ¤¥\
²«¬-
def num(str1):
a = b = c = d = 0
for i in str1:
if i.isdigit():
a += 1

elif i.isalpha():
b += 1

elif i.isspace():
c += 1
else:
d += 1
print('{%d\²{á%d\²%d\²Ìÿ{s%d\'% (a,b,c,d))
n u m (in p u t('Ɔ ¡¤¥E'))

# 5. ²¬-\{^Gëƻ\{
def compare_bigger(a,b):
if(a>b):
return a
else:return b
a= in t(in p u t("Ɔ ¡aÅE"))
b = in t(in p u t("Ɔ ¡bÅE"))
bigger=compare_bigger(a,b)
p rin t('GëÅf:',b igge r)

# 6. ²r§5\{²ƛY«¬-
def sum1(*args):
sum=0

for i in args:
sum+=i
return sum

p rin t('ƛY·¯E% d '% su m 1 (1,2,3,4))

7. ²Ȏ¡^5\ğbŋf-Ű*+²®{s|wŔȳwĬzw
JĒĩc²
dò\ğò\ŔŲƄ¡ĹÜǴargsȠ(²ö®Ȏ¡\
[1, 2, 3] (1 0, 2 0c
# ŀƎargsfb1,2,3,1 0,2 0)
def sum(*args):
li=[]
for i in args:
for j in i:
li.append(j)
return li
str1=sum([1,2,3],(10,20))
print(str1)

8. ²Ȏ¡^5\ğbŋf{ĭc²dò\ğò\ŔŲƄ
¡Ĺ
# ÜǴkw argsȠ(²
# ö®Ȏ¡\{'o n e ':1 } {'tw o ':2 }, ŀƎkw argsf{'o n e ': 1, 'tw o ': 2 }
def sum1(**kwargs):
return kwargs
str=sum1(**{'one':1},**{'two':2})
print(str)

òƄȶŤ¿²ȷȸõpȹęȶŤKȺ²ĔikE#ŋe²tË
*ƵąƘȻKȺķŝĹŵȼȽ²ȾKȺŀȿ²ȾKȺŀɀĩ
ğȶŤKȺȹęfįÓ²t{ĭµVzhȶŤKȺɁƚ
ò\õɂM6ğ
ŀŝeSŀŵe²-ąƘ«Ĭ
pğnłA{ĭĳf ¡Å
¢·¯^²÷į#ŋe²ÌÿõpąɃRYeȵ²
ť³õȤRį²ƯưɄɅ Ɇwzɇ ɈɉƲ

def avg(**kwargs):
sum=0

for i in kwargs.values():
sum+=i

avge=sum/len(kwargs)
print('#ŋeE% d '% avge)
def sq(di):
print('ȻKȺķŝĹŵȼ]fE',sorted(di.item s(),key=lam bda item:item [1],
reverse=True))
def max(di):
for i in di:

di1=sorted(di.items(),key=lambda item:item[1],reverse=True)
print('KȺŀŝE',di1[0:1])
def min(di):
for i in di:
di2=sorted(di.items(),key=lambda item:item[1])
print('KȺŀŵE',di2[0:2])

info={'ɊN':80,'±Ê':60,'±ɋ':60,'ɌǞ':70}
avg(**info)
sq(info)
max(info)
min(info)

1.ûò\\Ɂƚȃȫ²pƛ²òƄǥ6ǅ²õpd
Rĉ²
# ÖĜɁ ¢
# ƆM6strftim e ()ƮħŒȂȃÖeȄREfn.__n am e __
import time
def fn(func):
def inner(*args,**kwargs):
print('RĉE',fn.__nam e__)
print('ÖĜE',tim e.strftim e("% Y-% M -% d % H:% M:
%S",time.localtime(time.time())))
ret=func(*args,**kwargs)
return ret
return inner
@fn
def target_func():
print("KɁƚ!")
target_func()

2. QȩØ²ÆòƄź9ƂĻǅÈɍ6Į ¡6ĮRYƝ_²û6ĮN
ƄƠÙ²
# ƙƚKŭŃńƂĻ
register_username='admin'
register_password='123456'
def login(func):
def inner(*args,**kwargs):
for i in range(0, 3):
user_nam e = input("Ɔ ¡6ĮRE")
user_passw ord = input("Ɔ ¡Ɲ_E")
if user_name == register_username and user_password ==
register_password:
print("ƙƚK²ơƢ% s!" % u se r_n am e)
ret=func(*args,**kwargs)
return ret
break
else:
if i == 2:
print('ƙƚƧƨ²ƆÊƩƪƙƚ!')
else:
print('ƙƚƣƤ²Ƹũą%dƄƠÙ!' % (3 - i - 1))
else:
print('ƙƚƣƤų')
return inner

@login
def target_func():
print("ɎɏKŃńƂĻų")
target_func()

# 3. QȩØ²f5\Ȓƙƚľƶ² b6Įą3ƄƙƚƠÙc²
# pƛ²®¯ƙƚKį²¿Ǖmn6ƪƙƚį
flag=False
def login(f):
def inner(*args,**kwargs):
global flag
if flag:
ret=f(*args,**kwargs)
return ret

li=[]
i=0
while i<3:
i+=1

w ith open('6Į¸¹',m ode='r',encoding='utf-8',) as user:

user_nam e = input("Ɔ ¡6ĮRE")
user_passw ord = input("Ɔ ¡Ɲ_E")
for line in user:
li.append(line)
if user_name.strip()==li[0].strip() and
user_password.strip()==li[1].strip():
print('ƙǚK²ơƢ'+li[0])
ret=f(*args,**kwargs)
flag=True
return ret
else:
print('ƙƚƣƤ²ũɐ{}ƄƠÙ'.form at(3 - i))
else:
return 'ƙƚƣƤų'
return inner

@login
def add():
return 'ɑ'
@login
def delete():
return 'ɒ÷'
print(add())
print(delete())

4 ()*+98

4.1 3*+ 98

4.1.1 3 98

4.1.2 ÐTğö 99

# ÐT3
class Person:
# ŷɓO&
def __init__(self, name, age):
# ĿÓɔ%nam eYage
self.name = name
self.age = age
# O&
def say(self):
print (self.nam e + ‘ùEƥȿ’)
# P e rso n3ÐT*+1
p1 = Person(“±ɕ”, 2 0)
p1.say()
# Person3ÐT*+2
p2 = Person(“±ɖ”, 1 9)
p2.say()
±ɕùEƥȿ
±ɖùEƥȿ

4.1.3 3ɔ%ËÌŃń 100

tâãä æÃÄRYO&RąŞj.
_xxxE \âãä²ƾƃɗKŮ²Ħą3*+YŦ3*+ŃńāɘÃÄġ
__xxx__E ǅ¿ŋąâãä²ȹŞKŮġ
__xxxE əąKŮ²Ħą3*+ÛȮŃń²Ŧ3*+nÍrŃńĹā\KŮ

class Person:
__nam e = “±ɖ”
# 3KŮO&
def getname(self):
return self.__name
p = Person()
print(p.__name)
p.getname()
}ïEP yth o n^nýÆéïȒəąKŮ
ö®ȒɚÃÄŐĀ-tɛ6p._P e rso n __n am e>9Ńń

4.1.4 3^ȯĆO&Ëǥ6 102

ÐT3Ö6ÃÄµVzh*+ɜKŮĉfɔ%battrib u tec²6µVzh
*+9fKŮĉfKŮO&bm e th o dc²ɔ%YO&ȹĉf3KŮ
P yth o n^²-tĆ,(.)ɝĳsyŃń3KŮ
-tĆ,3RS3ğöÓ¿*+Rǥ6*Ĩɔ%SO&²ö®E
*+R.ɔ%Rw3R.ɔ%R
*+R.O&R()w3R.O&R()

# ÐT3
class Person:
info = 'This is a Person'

def __init__(self, name, age):
self.name = name

self.age = age
# KŮO&
def say(self):
print (self.nam e + ‘ùEƥȿ’)
@classmethod
def getinfo(cls)
print(cls.info)

4.1.5 əąɔ% 103

4.1.6 ŷɓO& 104

4.1.7 ƬŷO& 106

4.1.8 ć6¤çO& 107

ŦƕÁ 1

isinstanceEůÞ*+Ƶɔ3ťž3Sž3Ŧ3

issubclassEůÞ\3ťɞ\3Ŧ3

typ eE6üƮħ*+Ƶɔ3

4.2 ɟɠ5Ǵ 109

4.2.1 ɡưɟɠ 109

4.2.2 ®īɟɠɢ3 110

4.2.3 O&ɣ 111

4.2.4 su p e rO& 112

4.3 ŝŴĨ6 114

4.3.1 ¤çisinstancewissubclassYtyp e 114

4.3.2 3O& 115

3O&EM6@ classm e th o dǶȩO&²Ì1\3Ũbclsc

3O&²\ĢģpĽľȎ3²ĔÀì6cls

3O&b1c-tĆ,3RÍrǥ6ġ b2cĆ,3ğö*+ǥ6

3O&6Æ¾Ɨjava5\ŷɓıĲɤàüp yth o n3^Ħą\Ŀ
ÓO&²nȻɥnıĲĿÓ3

4.3.3 ɦǴO& 115

ɦǴO&EM6@ staticm e th o dǶȩO&²ÌŠą3O&^\3

ɦǴO&ǥ6E b1c-tĆ,3RÍrǥ6ġ b2cĆ,3ğö*+ǥ6

®¯6įstaticm e th o d²ƻưm-tɧɨā\se lf²ǧdā\O&ūK\ȯĆ
M6

3O&6Æ¾Ɨjava5\ŷɓıĲɤàüp yth o n3^Ħą\Ŀ
ÓO&²nȻɥnıĲĿÓ3

4.3.4 ÜǴ./3ğö 116

4.3.5 �slots� 117 ƨ<¦classğöɑɔ%

4.3.6 @property 118

4.3.7 Ŕ3 119

P yth o n ȠȊõ*+²ƻư*+mą*Ĩ“3(C lass)”²Sĉ“3 (typ e)”
P yth o n ^-t6 type(obj) yȇĹ*+“3”E

typ ev3 ũvtyp e²\ȏȣ3 

*+3 Ƒĳ3(class)²33 mĉĳŔ3vm e ta-class

4.3.8 ö¾V 120

ö¾Vǰę¾V^ŀ²ŀ¥ ǽ\¾V²ĹĦop;Ȍ
m-tǽ²Ï“ƃƶĦą*+ğö¾V”

4.3.9 ɩŦ3 122

ɩŦ3 bŘ Ed u ck typ in gcÆ]ǰę^ÜǴ3 »ɪÆā»ɪ
^²*+ąɫ ²nàɟɠÛ3Sğr½²ǧà"ūǅ
O&Yɔ%JĒ"ƅā\ R{y!üàɬɭɮ·ɯĘɭ·ɰǐ£¢ɩ
ŦţŤ² “ɩŦţŤ”-tāƱzɚE

“ūȆĹĦɱɲŉyɳɩŦwɴɵŉyɳɩŦwƑŉyÔɳɩŦ²ƻưāĦɱm-
tĈĉfɩŦ”

ÆɩŦ3 ^²}Æü*+9f²ĳɡưġǧn}*+Ƶɔ3 

*üǬǭp yth o n Iėyù²ň¸*üp yth o nɩŦ3 FGǬǭ²ƵǫɩŦ3
² ÆɶǡĘ^ǖƽ‘ÜǴ3 »ɪÆā»ɪ^²*+
ąɫ ²nàɟɠÛ3Sğr½²ǧà"ūǅO&Yɔ%
JĒ"ƅ’

4.4 ±· 123

4.5 L¿ÀÁ 123

ğƖ

# 1. d e f fu n c(*args): ȹęO&wwA n im al3*+\
from types import MethodType, FunctionType

class Animal:
def func1(self):
print('O&')
def func2(self):
print('O&')
@classmethod
def func3(cls):
print('O&')
@staticmethod
def func4():
print('')

def func(*args):
method_count=0
func_count=0
animal_count=0
for el in args:
if isinstance(el,MethodType):
method_count+=1
elif isinstance(el,FunctionType):
func_count+=1
elif type(el)==Animal:
animal_count+=1
else:

print('error!')
print('O&\E%d²\E%d²A n im al*+\E% d '%
(method_count,func_count,animal_count))
a1=Animal()
a2=Animal()
func(a1,a1,a1,a2,a2,a2,a1.func1,a1.func2,a1.func3,a1.func4,Animal.func3,Anim
al.func4)
print(type(a1.func1))
print(type(a1))
print(type(a1.func4))
print(type(Animal.func3))
print(type(Animal.func4))

2.C ircle3²pƛEikəąɔ%__rad iu s²ŷɓfɷɸÎÅ²ŷɓ
ĽľÅf0²
Ƭŷ ¢Ȱū¸¹²ȯĆO&Se tR ad iu s6üǰçɷɸ²ȯĆO&A re a¬-
ǎ(ɹ²¤çO&__str__6ü ¢ǎ(ɹ²
# ¤çO&__gt__6üFG\ǎ(ɹë±«ÐT\ğöeǩƖƶȒɚ
class Circle:
_ radius=0
def __init__(self,r=0):
self._radius=r
def __del__(self):
print('ɷɸf%dǎĈɺɻų'% self._radius)
def SetRadius(self,r):
self._radius = r
def Area(self):
return self._radius*self._radius*3.14
def __str__(self):
return str(self._radius*self._radius*3.14)
def __gt__(self, other):
return self.Area()>other.Area()
y1=Circle()
y2=Circle(1)
y1.SetRadius(2)
print(y1)
print(y2)
if y1>y2:
print('Sǎ1>Sǎ2 ')
else:
print('Sǎ1<Sǎ2 ')
del y1
del y2

3. ®âą\ƃɼɽɾ3In su ran ce P o licy
class InsurancePolicy:
def __init__(self, price_of_item):
self.price_of_insured_item = price_of_item
1cÐTŦ3EWĆɼV e h icle In su ran ce²ŷɓO&^Ć,su p e rdƃǁýþĹğ
öÃÄ^²
Ö6ğöÃÄve h icle _N am eýþWĆɿȪR{ÐT\ğöO&
ge t_ratebc²¬--ʀʁEƃǁ*0.0 0 1
2cÐTŦ3EʂʃɼH o m e In su ran ceŷɓO&^Ć,su p e rdƃǁýþĹğ
öÃÄ^²
Ö6ğöÃÄm e m b e r_n u m b e rýþʂʃKŮƘÐT\ğöO&
ge t_ratebc²¬--ʀʁEƃǁ*0.0 0 0 0 5

class InsurancePolicy:
def __init__(self, price_of_item):
self.price_of_insured_item = price_of_item

class VehicleInsurance(InsurancePolicy):
def __init__(self,price_of_insured_item,vehicle_Name):
super().__init__(price_of_insured_item)
self.vehicle_Name=vehicle_Name
def get_rate(self):
return self.price_of_insured_item*0.001

class HomeInsurance(InsurancePolicy):
def __init__(self,price_of_insured_item,member_number):
super().__init__(price_of_insured_item)
self.member_number=member_number
def get_rate(self):
return self.price_of_insured_item*0.00005

p 1 = V e h icle In su ran ce (1 0 0 0 0 0,'ʄʅ')
print(p1.get_rate())
p2=HomeInsurance(100000,4)
print(p2.get_rate())

4. ®âE
1c ʆ3Stu d e n t²ŷɓO&\nam eYyear²dĀƃýfğö
ÃÄÐT*+EJe rry²5ŒŴ
2cKȺ3G rad e²«Ëäɔ%Ep assin gf60ɑŷɓO&²Ć,
ȎÅûğöÃÄsco re
3cStu d e n t3ŷɓO&^²ÉÊ\KȺĬzgrad e sÆStu d e n t^ɑ
 \O&ad d _grad e (G rad e)²
¦O&opôÈƖƶȎ¡3 ťfG rad e²ĊƽȌ-tɑĹKȺĬ
z^²ť³nź9ɑɝĳ
4cÐT5\G rad e*+²KȺeǩf1 0 0,9 0,8 5,8 8,7 0e²«dĀĹJe rryK
ȺĬzȠ
# 5cG rad e^ɑis_passingO&²¬-¦KȺťË
# 6cStu d e n t^ɑ\¬-¦ʆ#ŋKȺO&ge t_ave rage ().
# 7cůÞJe rryťąnËKȺ
# 8c ¢Je rry#ŋKȺ
class Student:

def __init__(self,name,year):
self.name=name

self.year=year
self.grades = []
def add_grade(self,Grades):
if isinstance(Grades,Grade):
self.grades.append(Grades)
print('KȺǓɑų')
else:
print('error!')

def get_average(self):
sum=0

for el in self.grades:
sum+=el.score

self.average=sum/len(self.grades)
print('#ŋKȺ%.2 f'% se lf.ave rage)

stu 1 = Stu d e n t('Je rry','5ŒŴ')
p rin t('ɃRE% s,ŒŴE% s'% (stu 1.n am e,stu 1.ye ar))
class Grade:
passing=60
def __init__(self,score):
self.score=score

def is_passing(self):
if self.score<60:
print('nË')
else:
print('Ë')

g1=Grade(100)
g1.is_passing()
g2=Grade(90)
g2.is_passing()
g3=Grade(85)
g3.is_passing()
g4=Grade(88)
g4.is_passing()
g5=Grade(70)
g5.is_passing()

stu1.add_grade(g1)
stu1.add_grade(g2)
stu1.add_grade(g3)
stu1.add_grade(g4)
stu1.add_grade(g5)

stu1.get_average()

ʇÀ

1 2 3

class Person:
info='Class Person'
def __init__(self,name,age):
self.name=name
self.age=age
@staticmethod
def getinfo():
print('Class Person')
p=Person('x',16)
p.getinfo()
Person.getinfo()

def show(self):
print('ɃR:',se lf.n am e)
print('Œʈ',self.age)
def __showage(self):
print('Œʈ',self.age)
p = P e rso n ('ǒƘʉ',16)
#

p.show()
p.__showage()
Person.show(p)

#info='this is a person'
#__name='x'
def __init__(self,name,age,value):
self.name=name
self.age=age
self.__value=value
@property
def age(self):
print('get age')
return self._age
@age.setter
def age(self,age):
print('set age')
self._age=age
def setvalue(self,v):
self.__value=v

def __get(self):
return self.__value
def __set(self,v):
self.__value=v
def __del(self):
del self.__value
value=property(__get,__set,__del)
def show(self):
print(self.__value)
p=Person('x',16,1)
#p.setvalue(17)
p.show()
p.value=18
p.show()

p=Person('x',16)
p.age=17
print(p.age)
Person.gender='unknow'
print(Person.gender)
p=Person('x',16)
#

p.gender='male'
print(p.gender)
print(Person.gender)

#p=Person()
}ïEP yth o n^nýÆéïȒəąKŮ
ö®ȒɚÃÄŐĀ-tɛ6p._P e rso n __n am e>9Ńń
#print(p.__name)#AttributeError: 'Person' object has no attribute '__name'
#print(p._Person__name)
#nĦğö²*+Ô-tǥ63ɔ%
print(Person.info)
print(p.info)

4ɟɠ

class Person:

def __init__(self,name,age):
self.name=name

self.age=age
def getname(self):
print(self.name)
def printinfo(self):
print('ɃRE% s'% se lf.n am e)
class Student(Person):
def __init__(self,name,age):
super().__init__(name,age)
def printinfo(self):
print(self.nam e+' ʆ')
class Teacher(Person):
def __init__(self,name,age):
Person.__init__(self,name,age)
def teainfo(self):
print(self.nam e+'ʊȸ')
def printpersoninfo(person):
person.printinfo()
p=Person('a',17)
# p.getname()
# Person('w',18).getname()
stu=Student('s',12)
# Student('s',13).getname()
# stu.stuinfo()
tea=Teacher('t',14)
# tea.teainfo()
printpersoninfo(p)
printpersoninfo(stu)

new del str

#new O&
class Student:
_instance=None
def __new__(cls, *args, **kwargs):
if not cls._instance:
cls._instance=object.__new__(cls)
return cls._instance
def sleep(self):
print('sleeping...')
stu1=Student()
stu2=Student()
#

print(id(stu1),id(stu2))
print(stu1 is stu2)
#delO&
# class Person:
# def __del__(self):
# p rin t('d e lO&')
# p=Person()
# t=Person()
# del p
# t.__del__()
# strO&
class Student:

def __init__(self,name,age):
self.name=name

self.age=age
def __str__(self):
return str(self.__class__)+str(self.name)+','+str(self.age)
__ repr__=__str__
stu=Student('x',12)
print(stu)

5]ĬwJĒY{ĭ 127

5.1 -Ã3 n-Ã3 127

ɡư-Ű3

ŰP yth o nŀ0ë²ŃńåĒ3 ŔŲ»OVĆ,fo rſ
C-t6üƀƁ-Ű*+²P yth o n^-Ű3 ąE{s|²Ĭz²Ŕå²J
ĒY{ĭ²-tM6isinstance()yůÞ*+ť-Ű*+

-ʋʌn-ʋʌ

*+Ĉĉf-ʋʌ ²Ģģą\ʋʌÅ²ā\ÅÆ°\ʆßʍȈõnÙÃ
Ó²ǧİĢģ-t>9ňĩFG*ü Python ¤T3 yù²n-Ã3 õ
-ʋʌ² ®{s|²-Ã3 EĬzw{ĭwJĒ²ÿĀÆ×ÃÅÖʎŠą
×Ãid,ɧ&àșȚĝÅʏ%,Ūǧɧ&ʋʌ

5.2 Ĭz 128

5.2.1 ÐTĬz 128

ĦopȘdleʐªȵM6[]kŉyÏ-ÐTĬzö®E
list1=['Ɋʑ','ʒʓ','Ɍʔ','ʕñ']
p rin t(list1) # ¢f['Ɋʑ','ʒʓ','Ɍʔ','ʕñ']

Ĭzɝĳ

sum(iterable[, start]) �

ķ start ĿÛĸ)ĺ* Ĭz^ȵƛY«¬-ƐęÅ start Ľľf 0 Ĭz
ȵĆćf{² ĿÅ³nʖʗf{s|ö®:
list=[1,2,3,4]
print(sum (list)) # ¢·¯f10

sorted(iterable, key=None, reverse=False) �

¦*Ƶą-Ű*+>9ȼ]ɝĳkeyrƾ\²ā\Ħrƾ
\ŔŲ²ĽľfN one²reverse\ĤĥÅ®¯ǰçfTru e²ĬzŔŲdĈʘ
]ȼĬ²ĽľfFalse

zip([iterable, ...])
zip () 6üdĪï5\-Ű*+ĳf²d*+^*ĨŔŲʙiKŔå²
ȓ¿¬-àāɘŔååK-Űzip*+²-tM6 list() ´čy ¢Ĭz
alist=[1,2,3]
blist=[4,5,6]
clist=[6,7,8,9,10]
zip p e d = zip (alist,b list) # ¬-*+
l3 = list(zipped)
p rin t(l3) # list() ´čfĬz²·¯fE[(1, 4), (2, 5), (3, 6)]
p rin t(list(zip (alist, clist))) # ŔŲ\ŀȦĬzƷ [(1, 6), (2, 7), (3, 8)]
a1, a2 = zip (*l3) #  zip ňʚ²zip (*) -ǽfʛ²¬-ďɶʜʝV
list(a1) # ¢·¯fE(1, 2, 3)
list(a2) # ¢·¯fE(4, 5, 6)

Ĭz3 ɝĳO&

stu List= ['Ɋʑ','ʒʓ','Ɍʔ']²>9®âĬzɝĳ

}ï:
āȠso rtO&YĻǖ3 ¤çso rte dn²so rte dnÙ×ÃǷyĬz
¤¥²ǧ¬-\ĄĬz*+²list.sortO&Ù×ÃǷĬz¤¥

ʞʟʠÙfĄ*+ ʡýþ²Ɗf*+Ũ ʡ²*+^ŔŲʢY
Ƿ*+ɔü\u6
ʞʟʠ¿®¯×ÃǷ*+^n-Ã3 ŔŲ²ʣȒʤʆį\Ąn-Ã
ŔŲ²ʟʠ*+^ŔŲʢž)Ƿyn-ÃŔŲ²ƵtŠąIʆ×Ã®¯×Ã
Ƿ*+^-Ã3 ŔŲ²àü-ÃŔŲÆʟʠ*+^Šą ʡĄýþ²Ƿ
+Yʟʠ+^-ÃŔŲ*+²Ƶtʟʠ*+ÔÙ×Ã

ɤ ʥʟʠ copy.deepcopy(d)

5.2.2 ʦrĬz 129

5.2.3 Ğ<ĬzŔŲ 130

5.2.4 ȲȜĬzŔŲ 131

5.2.5 ʧ¡ĬzŔŲ 132

5.2.6 ɒ÷ĬzŔŲ 132

5.2.7 ƮħĬz^ŀëŔŲ 133

5.2.8 ƮħĬz^ŀ±ŔŲ 133

5.2.9 ȹęŔŲ¢Ƅ 133

5.2.1 0 ęĬzłA 134

5.2.1 1 Ĭz^ŔŲȼ] 134

5.3 Ŕå 135

5.3.1 ÐTŔå 136

Ŕån-Ã]Ĭ3 ²-tdŔåȆƒĦŔŲĬz²n>9wǶ×
Yɒ÷ŔŲɝĳ

ŔåM6leº6±klkŉyåª²ŔåĬz3ʨ²nŔ
åÅn-×Ã²ŔåÐT®âƵhE

Ć,tu p leÔ-tŷTŔå²¦®âE
tuple(value)
-tdĬzw{ĭSė{s|´čfŔå²ö®E
list1=[1,2,3,4] tuple(list1) #·¯fE(1,2,3,4)
str= 'w jz w j' p rin t(tu p le (str)) #·¯fE('w ','j','z',' ','w ','j')

ŔåŃń

tup1 = ('Taobao', 'Jingdong', 1, 2)
tup2 = (1,2,3,4)
p rin t(tu p 1 [0]) # ¢·¯fE 'Taobao'
p rin t(tu p 2 [1:3]) # ¢·¯fE (2, 3)

5.3.2 ÐTȪą\ŔŲŔå 136

5.3.3 ʦrŔå 137

tup1 = (12, 34.56)
tup2 = ('abc', 'xyz')
tu p 3 = tu p 1 + tu p 2 # ÐT\ĄŔå
p rin t (tu p 3) # ¢·¯fE(1 2, 3 4.5 6, 'ab c', 'xyz')

5.3.4 ƮħŔå^ŀëŔŲ 137

5.3.5 ƮħŔå^ŀ±ŔŲ 137

5.3.6 ŔånÃ% 137

5.4 JĒ 138

5.4.1 ÐTJĒ 138

-tM6ëkl { } Sė set() ÐTJĒ²}ïEÐT\JĒĢģ6 set() ǧ
n { }²Ūf { } 6yÐT\{ĭJĒ^ŔŲĢģ-ʋʌn-Ã
3 õ -ʋʌ

JĒŃńËȬĄ

àüJĒɧ]²Ƶt-tĆ,ƀƁȲȆJĒKŮSėĆ,inSė not inȱȲű
ȵŔŲťJĒ^KŮ

5.4.2 ŔŲʏ% 138

5.4.3 ʧ¡JĒŔŲ 138

5.4.4 WJ 139

5.4.5 «J 139

5.4.6 ɀJ 140

5.4.7 *ĉɀJ 140

5.4.8 ŦJ 140

5.4.9 ɢJ 141

5.5 {ĭ 141

5.5.1 ÐT{ĭ 141

{ĭP yth o n ^ʏʩʪ3 ²6üýþȪąʫª{ĭ*+-
t¥ʬĪï\P yth o n*+
{ĭ3 Y]Ĭ3 êǩýþYŃńªOVn
{ĭè-tĪīÌÿ*+²ǧ]ĬĦM6{bļucĳfè²{ĭ^
èɧ]

5.5.2 ĿÓ{ĭ^ŔŲ 141

5.5.3 Ƕ×/ʧ¡{ĭŔŲ 142

5.5.4 ɒ÷{ĭ^ŔŲ 143

5.5.5 ę{ĭ^ŔŲ\ 144

5.5.6 ʭ÷{ĭ^ƵąŔŲ 144

5.5.7 ůÞ{ĭ^ťýÆè 144

5.5.8 ʦr\{ĭ 145

5.5.9 Ʈħ{ĭ^èJĒ 146

5.5.1 0 Ʈħ{ĭ^ÅJĒ 146

5.5.1 1 Ʈħ{ĭ^ŔŲå 146

5.5.1 2 ʞʟʠ 147

5.5.1 3 ʥʟʠ 148

5.6 ŝŴĨ6 149

5.6.1 Ȋʮ 149

5.6.2 ĬzʆKzȧV 149

5.6.3 ʆKØ 150

Æ Python ^²M6į yield ĈĉfʆKØbge n e rato rc
ȉȯĆn²ʆKØ\¬-ŰØ²Ħ6üŰɝĳ²
ȬǽʆKØm\ŰØ
Æǥ6ʆKØ?9,^²òƄʯĹ yield ÖÙʰʱ«ƃýūǅƵą
?9¸¹²¬- yield Å, «ÆâƄź9 next() O&ÖķūǅĝçɟǕ?9
ǥ6\ʆKØ²¬-\ŰØ*+

d e f fib o n acci(n): # ʆKØ - ƹƺƻƼ
a, b, counter = 0, 1, 0
while True:
if (counter > n):
return

yield a
a, b = b, a + b
counter += 1
f = fib o n acci(1 0) # f \ŰØ²àʆKØ¬-ʆK
while True:
try:
print (next(f), end=" ")
except StopIteration:
exit()
¢E
0 1 1 2 3 5 8 13 21 34 55

5.6.4 ŰØ 151

ŰP yth o nŀ0ë²ŃńJĒŔŲ»OVĆ,fo rſC
-t6üƀƁ]ĬwJĒw{ĭĩ-Ű*+²Ô-tM6ŰØy>9JĒƀ
Ɓ

ŰØ\-tɁʲƀƁĝç*+ŰØ*+ķJĒ\ŔŲ
ĿŃń²ÍĹƵąŔŲĈŃńƔ·ŅŰØĦʳǅnÙ¿ȔŰØą\
O&Eiter() Y next(){s|²ĬzSŔå*+õ-6üÐTŰØE

{s|²ĬzSŔå*+õɔü-Ű*+²ƊnŰØ²-tM6
isinstanceO&ůÞ*+ť-Ű*+

Ș\3ĳf\ŰØM6opÆ3^ğ\O& __iter__()  __next__() 
__ ite r__() O&¬-\ŞŰØ*+² ā\ŰØ*+ğį __next__() O
&«Ć, StopIteration ŧćĻŰƔK
__n e xt__() O&bP yth o n 2 Ƞ next()cÙ¬-â\ŰØ*+
ÐT\¬-{ŰØ²ĿÅf 1²ʴņȏ 1E

#\ʆKƹƺƻƼĬŰØE
from collections.abc import Iterable,Iterator
class Fibonacci:
def __init__(self):
self.a=0
self.b=1
def __next__(self):
tmp=self.a
self.a=self.b

self.b=tmp+self.b
return self.b
def __iter__(self):
return self
fibs=Fibonacci()
print(isinstance(fibs,Iterator))
for i in range(6):
print(next(fibs))

5.7 ±· 153

5.8 L¿ÀÁ 153

ğƖ

1. dStu d e n tIn fo^¸¹t{ĭµVýþÆĬzȠ
¢E
[{' l': '1445204009', 'ɃR': 'Ɍʵ', '#ÖKȺ': '100', 'ȈʶKȺ': '90'},
{' l': '1445204013', 'ɃR': 'ǿʷ', '#ÖKȺ': '95', 'ȈʶKȺ': '67'}
...
#]
lst=[]
with open('StudentInfo.csv',mode='r',encoding="gbk") as f:
line=f.readline().strip()
title=line.split(',')
for line in f:
dic={}
line=line.strip()
data=line.split(',')
for i in range(len(title)):
dic[title[i]]=data[i]
lst.append(dic)
print(lst)

# ¡\{²´čK^Z{F®E1 2 3 4 5 6 7 8 9 0 -> ǙǈǆǌǊǚǛǜǝǘ
lst1=[1,2,3,4,5,6,7,8,9,0]
lst2=['Ǚ','ǈ','ǆ','ǌ','Ǌ','ǚ','Ǜ','ǜ','ǝ','ǘ']
n u m b e r= e val(in p u t('Ɔ ¡\{E'))
for i in lst1:
if number==i:
index=i-1
p rin t('´č¿^Z{E',lst2[index])

Ɩƶ_ĔikɘÕƠʤʆ{Ssl²ƆğÕƠʆKå6ĝƖƶ_

# ò\{s-të{áw±{áS{²ąİĦāN»3 ^»
import random
list1=[]
for i in range(10):
list1.append(str(i))
for i in range(65,91):
list1.append(chr(i))
for i in range(97,123):
list1.append(chr(i))
code=random.sample(list1,6)
for i in code:

print(i,end='')

ą®âʸʹʺE568²239²368²425²121²219²834²1263²26²
Ɔ ¡Õï\ʺê>9ʸʹʻŌ²«ʼ*ʻŌ¢ʸʹ>9ķëĹ±
Yķ±Ĺë>9ȼ]²
# «ƛ¢ā\êʸʹ#ŋʺ
prices=[568,239,368,425,121,219,834,1263,26]
lst1=[]
p rin t('ʸʹʺƐeĤE',prices)
p rin t('Ɔ ¡\Õïʸʹʺê')
m in _p rice = e val(in p u t('ƆÈ ¡êŀ±ÅE'))
m ax_p rice = e val(in p u t('Ɔ ¡êŀëÅE'))
for j in prices:
if max_price>=j>=min_price:
lst1.append(j)
print(lst1)
lst2=sorted(lst1,reverse=True)
lst3=sorted(lst1)
p rin t('ķëĹ±ȼ]E',lst2)
p rin t('ķ±Ĺëȼ]E',lst3)
sum=0
for i in lst1:
sum+=i
ave=sum/len(lst1)
p rin t('¦ê#ŋʺE',ave)

d ʆ*+ý¡Ĭz^²«ȻKȺ* ʆ>9ȼ], «ƮħKȺŀŝYKȺŀŵ
ʆ¸¹²
«dŀŝeYŀŵe ʆķĬzɒ÷²ŀ¿ƪ*Ĭz>9ʟʠ²*ʟʠĬz
>9ʽ´ ¢
import random
students=[]
class Student:

def __init__(self,name,age,score):
self.name=name
self.age=age
self.score=score
def info(self):
print(self.name,self.age,self.score)
for i in range(5):
age=random.randint(18,21)
score=random.randint(0,100)
students.append(Student('ɊN' + str(i)+'l', age, score))
p rin t(' ¢ǷĿªE')
for s in students:
s.info()
students.sort(key=lambda stu:stu.score,reverse=True)
p rin t(' ¢ȻKȺʾ]ȼ]E')
for i in students:
i.info()
p rin t(' ¢KȺŀŵ ʆE')
lowest=students.pop(-1)
lowest.info()
p rin t(' ¢KȺŀŝ ʆE')
highest=students.pop(0)
highest.info()
p rin t(' ¢dŀŝeYŀŵeɒ÷¿ ʆĬzbʾ]ȼĬcE')
for i in students:
i.info()
p rin t('*ɒ÷¿ ʆĬz>9ʟʠʽ´ ¢E')
stuList=students.copy()
stuList.reverse()
for i in stuList:
i.info()

Ɔǰę\ȿʿˀǽȹ²ò\õ*Ĩ\]l²6Į-Úª£h“Ɔ ¡Ƹ
Ōȵ”Ōˁ]lź9ňĨɝĳ²ikE
b1cɑȿʿE6ĮÚª£h“Ɔ ¡pɑȿʿE” ¡pɑȿʿɃR²
ɑ¿Ù£h“ȿʿɑK”
b2cɒ÷ȿʿE6ĮÚª£h“Ɔ ¡ɒ÷ȿʿɃRE” ¡pɒ÷ȿʿɃR²
ɒ÷¿£h“ɒ÷K”
b3cǗ}ȿʿE6ĮÚª£h“Ɔ ¡pǶ×ȿʿɃRE”Y“Ɔ ¡Ƕ×¿
ȿʿɃRE”
eǩ ¡Ƕ×ǅYǶ×¿ȿʿɃR²Ƕ×¿Ù£h“Ǘ}K”
b4c/hȿʿEŸ6ĮũŠąɑ,ȿʿ²£h“ȿʿĬzf”²ť³¬-ò\
ȿʿɃR
# b5cȔ¢EȨȿʿȹ
friend=[]
p rin t('1.ɑȿʿ\n 2.ɒ÷ȿʿ\n 3.Ǘ}ȿʿ\n 4./hȿʿ\n 5.Ȕ¢\n ')
while True:
option = int(input('Ɔ ¡ƸŌȵE'))
if option==1:
add_nam e=input('Ɔ ¡pɑȿʿɃRE')
friend.append(add_name)
print('ȿʿɑK!')

elif option==2:
del_nam e=input('Ɔ ¡ɒ÷ȿʿɃRE')
friend.remove(del_name)
print('ɒ÷K!')
elif option==3:
before_nam e=input('Ɔ ¡pǶ×ȿʿɃRE')
after_nam e=input('Ɔ ¡Ƕ×¿ȿʿɃRE')
index=friend.index(before_name)
friend[index]=after_name
# friend.remove(before_name)
# friend.append(after_name)
print('Ǘ}K')
elif option==4:
if len(friend)==0:
print('ȿʿĬzf')
else:
for i in friend:
print('ȿʿĬzE',i)
else:
exit()

ÊÊǪÆ ˂^Ɔɘ ŉƒȵń˃ǥȲ,fįğƖ˄˅%,
ÿÈ6ęƠʆKįN \1Ĺ1000ÕƠ°(N ˆ1 0 0 0),
*üÌ^ɣĞ{,Ħƃˇ\,ȘÌˈňˉˊ,n*Ĩăn
ʆ l
ȓ¿ƪȘāɘķëĹ±ȼ],ȻɥȼȿĶ]ˉȜ ƒǥȲ
ƆƥˋˌÊÊƔK“ˉɣ”“ȼ]”ɿĳ(\ţŤ6öȠ-Ùą5åª,ʌ
ˍëʂĊƽǼǽ)
'''

è ;E
1). ʆKįN \1Ĺ1000ÕƠ°(N ˆ1000)
2). ˉɣ: Ì^ɣĞ{,Ħƃˇ\,ȘÌˈňˉˊ
3). ķëĹ±ȼ]
'''

import random
# 2). ˉɣ: Ì^ɣĞ{,Ħƃˇ\,ȘÌˈňˉˊ.ʆK\JĒ
nums = set()
N = in t(in p u t('Ɔ ¡ÕƠ°N \: '))
# 1). ʆKįN \1Ĺ1000ÕƠ°(N ˆ1000)
for count in range(N):
num = random.randint(1, 1000)
nums.add(num)
# 3). ķëĹ±ȼ], li.sort()ˎ*Ĭz>9ȼ]ġ sorted()O&-t*Īïª3
ȼ]
print(sorted(nums, reverse=True))

ğ ʆ¸¹ˀǽȹ^ɑ ʆ²«ȻɥKȺķŝĹŵ>9ȼ]²ŀ¿
* ʆ¸¹>9 ¢
#b1cɑ ʆ¸¹
#M6ĬzY{ĭ* ʆ¸¹>9ýþ
students = []
while True:
nam e = input("Ɔ ¡R{E")
if not nam e: # ūR{fÖ²·ŅſC
break
try:
age = int(input("Ɔ ¡ŒʈE"))
score = int(input("Ɔ ¡KȺE"))
except:
print(" ¡n°µÅ²ƆɣĄ ¡")
info = {"name":name,"age":age,"score":score}
students.append(info)
p rin t(" ʆ¸¹ƚ¡ƔƦųųų")
#b2c* ʆķŝĹŵ>9ȼ]
p rin t("Ȼ ʆKȺŝˏŵgh")
sortedStudent = sorted(students,key=lambda x:x.get('score'),reverse=True)
print(sortedStudent)
# b3cƛKȺ#ŋÅ
# M6ĬzʆKzȧVƮħKȺĬ²ȓ¿ƛȇƐKȺYƐƘ²ŀƎƛȇŋÅ
g=[lst['score'] for lst in sortedStudent]
p rin t("KȺĬzfE")
print(g)
p rin t("KȺ#ŋÅfE")
print(sum(g)/len(g))
#b4c* ʆ¸¹>9 ¢
def show_student(students):
if not students:
print("ɧª¸¹ːːːːː")
return

print("R{".center(8),"Œʈ".center(4),"KȺ".center(4))
for info in students:
print(info.get("name").center(10),str(info.get("age")).center(6),str(info.get("
score")).center(6))
show_student(sortedStudent)

ûōZ²ö®E“w h o h ave an ap p le ap p le is fre e fre e is m o n e y yo u
know ”²Ɔȹęó¢Ƅ
import re
str1='who have an apple apple is free free is money you know?'\
'I have an apple, and what about you?'
str2=re.split(r'[\d\W\s]',str1)
print(str2)
#ˉɣĞȵ
set1=set(str2)
print(set1)
#ȘJĒ´fĬz²JĒŔŲŠąĶ]²Šąļuɔ%
list1=list(set1)
print (list1)
dir1 = {}
for x in range(len(list1)):
if x=='':
pass
else:
dir1[list1[x]] = 0 #{ĭÅĿf0
for y in range(len(str2)):
if list1[x] == str2[y]:
dir1[list1[x]] += 1
print (dir1)

#* ʆÕƠŌˁ>¡m athYP yth o nˑŴ²ȇĹ˒Æm athˑ˓Æp yth o nˑ ʆ
JĒ²
Æp yth o nˑƊnÆm athˑ ʆJĒ²
Æ ˑSėp yth o nˑ²ƊŠąÖŭ\ˑŴ ʆ²
tË>¡m athˑSėp yth o nˑ ʆJĒ
stu List= {'Ɋʑ','ʒʓ','Ɍʔ','ʕñ','˔˕','ʍÊ','˖˗','˘'}
mathSet=stuList.copy()
pythonStu=stuList.copy()
#ÕƠeˑ
mathClass1=set()
pythonClass2=set()
for i in range(4):
mathClass1.add(mathSet.pop())
p rin t('m athˑ ʆE\n ',m ath C lass1)
for i in range(5):
pythonClass2.add(pythonStu.pop())
p rin t('p yth o nˑ ʆE\n ',p yth o n C lass2)

#bWJc
mathAndPython=mathClass1.intersection(pythonClass2)
p rin t('˒Æ ˑ˓Æp yth o nˑ ʆ:\n ',m ath A n d P yth o n)
#bɀJc
pythonNotMatn=pythonClass2.difference(mathClass1)
p rin t('Æp yth o nˑƊnÆm athˑ ʆ:\n ',p yth o n N o tM atn)
#bǲJc
notMath=stuList.symmetric_difference(mathClass1)
p rin t('Æ ˑSėp yth o nˑ²ƊŠąÖŭ\ˑŴ ʆ:\n ',n o tM ath)
#b«Jc
pythonOrMath=mathClass1.union(pythonClass2)
p rin t('>¡ ˑSėp yth o nˑ ʆ:\n ',p yth o n O rM ath)

6 {s| 157

6.1 {s|ć6ɝĳ 157

Ńń{s|

ÃÄv[ĿļuE·ŅļuEņł]
nij·ŅĝçâĻ*Ĩª²Ċċŋ-
ņłŌħńÁ²Ċċŋ-²Ľľņłf1

{s|ĬzħÅĶ]E
ķ0)ĺ ĿSķ-1)ĸ Ŀ

Ȳ˙{s|

ÏȲȜŦ|Æ{s|^ĝçS¢Ƅ

1. fin d ()EȲȜű\Ŧ|ťijÆ¦{s|^²®¯Æ¬-ā\Ŧ| Ŀĝç
âĻ²ť³¬--1
&E
{s|.fin d (su b [,start[,e n d]])
Ì^E ĿĝçâĻ²·ŅĝçâĻ-tƇƈ²zhÆ°\{s|]Ĭ^ȲȜĽ
ľıĲâstartÅf0²endÅf{s|łA

str = 'Rose, rose, I love you. Rose, rose, I fallin love with you.'
print(str.find('rose')) #6
print(str.find('rose', 10, 20)) #-1
print(str.find('rose', 20, 40)) #30

2.in d e x()Eȱļű\Ŧ|ťijÆ¦{s|^²®¯Æ¬-ā\Ŧ| Ŀĝç
âĻ²ť³³ʀŧć
&E
{s|.in d e x(su b [,start[,e n d]])
rfind(sub[,start[,end]])EYfin d ()ň²ƊȲȜO)ķĺ Ŀ
rindex(sub[,start[,end]])EYindex()ň²ƊȲȜO)ķĺ Ŀ

str = 'Rose, rose, I love you. Rose, rose, I fallin love with you.'
print(str.index('rose')) #6
print(str.index('rose', 20, 40)) #30
print(str.index('rose', 10, 20)) #ValueError: substring not found

print(str.rfind('rose')) #30
print(str.rfind('rose', 10, 20)) #-1
print(str.rindex('rose')) #30

3. co u n t()E¬-ű\Ŧ|Æ{s|^¢Ƅ
&E
{s|.co u n t(su b [,start[,e n d]])

print(str.count('rose')) #2
print(str.count('rose', 10, 20)) #0
print(str.count('rose', 10)) #1

Ƕ×{s|

1. re p lace (): Ș{s|^o ld{s|ǵčKnew {s|²ŀƎ¬-ǵč¿Ą{s
|
&E
{s|.re p lace (o ld, n e w [,m ax])
}ïEǵčƄm ax®¯Ƨ¢Ŧ|¢Ƅ²³ǵčƄf¦Ŧ|¢Ƅ

print(str.replace('rose', 'python'))
print(str.replace('rose', 'python', 1))
print(str.replace('rose', 'python', 10))
p rin t(str) #Ƿ{s|Šą×Ã, {s|n-Ã3

new_str = str.replace('rose', 'python')
print(new_str)
#Rose, python, I love you. Rose, python, I fallin love with you.
Rose, python, I love you. Rose, rose, I fallin love with you.
Rose, python, I love you. Rose, python, I fallin love with you.
Rose, rose, I love you. Rose, rose, I fallin love with you.
Rose, python, I love you. Rose, python, I fallin love with you.

2. sp lit():Ȼɥžeºs*{s|>9Ȋʐ²¬-àȊʐ·¯åKĬz
&E
{s|.sp lit(se p, m axsp lit)
}ïE
se pzhžeºs²ĽľÅfN oneġ
m axsp litzheºs¢Ƅ²ĽľÅf-1²zhn*ȊʐƄƒƨ<ġ
sp lit()¬-Ĭząm axsp lit+ 1\ŔŲåK

list1 = str.split(',')
print(list1)
list2 = str.split(',', 2)
print(list2)
# ['Rose', ' rose', ' I love you. Rose', ' rose', ' I fallin love with you.']
# ['Rose', ' rose', ' I love you. Rose, rose, I fallin love with you.']

3. jo in ()Ed]Ĭ^ŔŲtž{sqrK\Ą{s|
&E
str.join(seq)
}ïEse q\]Ĭ*+²strM6qrs

list3 = ['I', 'love', 'Python']
print(' '.join(list3)) #I love Python
str1 = ','
print(str1.join(list3)) #I,love,Python
str2 = '...'

print(str2.join(list3)) #I...love...Python
str3 = '_'
print(str3.join(list3)) #I_love_Python

ë±´č
str.cap italize () #d{s|\{áë²Ìÿ{áõ±
str.title () #d{s|^ò\óô{áë
str.lo w e r() #d{s|^Ƶą{áõ±
str.u p p e r() #d{s|^Ƶą{áõë
str.sw ap case () #d{s|^Ƶą±{áÃKë²Ƶąë{áÃK±

*ŏOV
1. lju st()E¬-\dǷ{s|ĸ*ŏ²«M6ž{sd{s|˚˛ȅ*ĨłA
Ą{s|
&E
{s|.lju st(w id th [,fillch ar)
}ïEž{s|fillch arĽľf²®¯žłA±üǷ{s|łA²³¬-
Ƿ{s|
2. rjust(width[,fillchar)
¬-\dǷ{s|ĸ*ŏ²«M6ž{sd{s|˚˛ȅ*ĨłAĄ{s
|
3. center(width[,fillchar)
¬-\6fillch ar˚˛²ŎAfw id th²Ƿ{s|˜^Ą{s|

my_str = 'python'
print(my_str.ljust(10))
print(my_str.ljust(10,'.'))
# python
# python....
print(my_str.rjust(10,'.')) #....python
print(my_str.center(10,'.')) #..python..

ɒ÷
str.strip () #ɒ÷stræĕYŊĕ
str.lstrip () #ɒ÷stræĕ
str.rstrip () #ɒ÷strŊĕ

ůÞ{s|

ůÞO&¬-fĤĥ3 ²ÏTru e²False.

1. startsw ith () 6üȱȲ{s|Æ[start,e n d]Ł¤ťtžŦ{s| æ®¯
¬-Tru e²ť³¬-False
&E
{s|.startsw ith (su b str, b e g= 0, e n d = le n (strin g))
2. e n d sw ith () 6üȱȲ{s|Æ[start,e n d]Ł¤ťtžŦ{s|·Ŋ®¯
¬-Tru e²ť³¬-False
&E
{s|.e n d sw ith (su b str, b e g= 0, e n d = le n (strin g))

my_str = 'python java C python C++'
print(my_str.startswith('python')) #True
print(my_str.startswith('python', 10)) #False
print(my_str.endswith('C++')) #True

3. isalp h a()E®¯{s|ȅ˝ą\{s«İƵą{sõ{áS^Z{³¬-
Tru e, ť³¬- False
4. isd igit()E®¯{s|^Ħij{³¬- True² ť³¬- False²U n ico d e
{²b yte{b{Ĝc²ȗŜ{bx{Ĝc²nijǂ{{²˞˟{
5. isaln u m ()E®¯{s|ȅ˝ą\{s«İƵą{sõ{áS{³¬-
Tru e²ť³¬- False
6. issp ace ()®¯{s|^Ħijˠ²³¬- True²ť³¬- False.
7. isn u m e ric()®¯{s|^Ħij{²³¬- True²ť³¬- False²{-t
EU n ico d e {²ȗŜ{bx{Ĝc²˞˟{²ǂ{{nijb yte{
Ĝ

str1 = 'I love python!'
p rin t(str1.isalp h a()) #{s|^ą²ŪƉ¬-False

str2 = '1234'

p rin t(str2.isd igit()) # Tru e²ąffalse

str3 = '1234abcd'
p rin t(str3.isaln u m ()) # Tru e²ąffalse

str4 = ' '

print(str4.isspace()) #True
print(str1.isspace()) #False

str5 = "1 2 3 4ˡˢ"
print(str5.isnumeric()) #True
print(str5.isdigit()) #False

8. istitle ()²®¯{s|ĻÁÓ³¬- True²ť³¬- False
9. islo w e r()²®¯{s|^ijȅ˝\êeë±{s²«İƵąāɘ{sõ
±²³¬- True²ť³¬-False
1 0. isu p p e r()²®¯{s|^ijȅ˝\êeë±{s²«İƵąāɘ{s
õë²³¬- True²ť³¬-False

str1 = 'python'
str2 = 'PYTHON'

str3 = 'I Love Python'

print(str1.islower()) #True
print(str2.isupper()) #True
print(str3.istitle()) #True

6.1.1 ÐT{s| 157

6.1.2 ulwxulwNulêǩ 157

6.1.3 {s|FG 159

6.1.4 {s|Ȋʐ 160

6.1.5 {s|ȱļ 161

6.1.6 ǵč{s|^{s 161

6.1.7 ˉ÷{s| 162

6.1.8 Ğ<{s| 163

6.1.9 qr{s| 163

6.1.1 0 Ʈħ{s|łA 163

6.1.1 1 ë±´č 164

6.1.1 2 ţŤ{s|åKĕe 164

6.2 VÓO& 165

6.2.1 ěĝs 165

1. %s

p rin t('ŐR{E% s' % n am e)
2. f'zȧV'
print(f'ŐR{{n am e }²Ő lE{sn o }²ŐőŒ{age }œ')

6.2.2 fo rm atO& 165

M6{s|^fo rm atO&Ô-t>9{s|VÓɝĳ &Ć,
{} Y : yǵtǅ %.

&E
str.form at(*args²**kw args)
str{s|à“”Y“{}”åK² “{}”ĳ6Y% ň²6y¶<Ƕ×{s|^ʧ¡Åĝ
ç
®¯¾ˣ{s|^ą5\“{}”²«İ“{}”¤ŠąžĪī]lbķ0 Ŀ lc²³Ľ
ľȻɥ“{}”¢Ķ]eǩ6>9ǵč
®¯¾ˣ{s|^“{}”ÊƽM6į]l²³opȻɥ]l*Ĩ>9ǵ
č
}ïEfo rm atO&¬-VÓ{s|ˤ²«nÙ×ÃǷ{s|¤¥

# 1. Ć,ĝç
str1 = '{0 } l{1 },{0 }Œʈ{2 }'
p rin t(str1.fo rm at('ʒ˥', 1002, 22))

# 2. Ć,è{Rĉ
str2 = '{n am e } l{sn o },{n am e }Œʈ{age }'
p rin t(str2.fo rm at(n am e = 'ɊN', sno = 1001, age = 20))
fo rm atO&¬-VÓ{s|ˤ²«nÙ×ÃǷ{s|¤¥

# 3. Ć,*+ɔ%
class Student:

def __init__(self, name, sno, age):
self.name = name
self.sno = sno
self.age = age

s = Stu d e n t('±Ê', 2001, 20)
m y_str = '{stu.n am e } l{stu.sn o },{stu.n am e }Œʈ{stu.age }'
p rin t(m y_str.fo rm at(stu = s)) #±Ê l2001,±ÊŒʈ20

6.3 Ċ³zȧV 166

6.3.1 Â & 166

ÆğǸĨ6^²ÆM6ȯĆ{sÖ²opƒĚƽ˦ú²®¯M6Ş{s³
˦ú¾V6ü¾˧˦ú

[0-9]: ˦úĪ{²0~9
[a-z]: ˦úĪ±{s
[A -Z]E˦úĪë{s
[0-9A -Za-z]E˦úĪ{Y{á
[0-9A -Za-z_]E˦úĪ{w{áYâãä

Ş{s

Ć6˦ú

.*bǦcEÌ^.bc-t˦úĪï{sb÷č9sc²*bǦcz˦úǅ(
{sɧƨƄ²ƵtĀåĒÆŉm-t˦úĪï{sį
.*?bǦńcE˨-˝˦úĪï{s

#˩˪ǹ˩˪˦ú
content = 'Hello 1234567 World_This is a Regex Demo'
re_pat1 = re.compile(r'^he.*(\d+).*Demo$', re.I)
re_pat2 = re.compile(r'^he.*?(\d+).*Demo$', re.I)
p rin t(re _p at1.fin d all(co n te n t)) # ['7 ']² Æ˩˪˦úâ².*Ù˦ú˨-5{
s
p rin t(re _p at2.fin d all(co n te n t)) # ['1 2 3 4 5 6 7 ']²ǹ˩˪˦úm˨-˦ú˝{
s

6.3.2 re¾ M6 168

P yth o n^ re ¾ £˫įĊ³zȧV²¦¾ £˫Z˦úȲȜwZǵčwZ
eºĩ

flagsȪ®âzƵhE

re¾ ć6E

compile()

1. compile()
d\{s|µVĊ³zȧV K\Ċ³zȧVP atte rn*+tˬüɣ
ĞM6Ċ³zȧV²˭˝ȹ ɺ
co m p ile &V®âE
compile(pattern,flags=0)
p atte rn\Ċ³zȧV²flagsž˦ú¾V²ˮƈë±²59¾Vĩ

import re
content = 'I like Python , I am in Python class'
co m R e ge x = re.co m p ile (r'P yth o \w ', re.I) # nêeë±
p rin t(typ e (co m R e ge x)) # p atte rn3
alist = comRegex.findall(content)
print(alist) #['Python', 'Python']

re_phone = re.compile(r'\d{3}-\d{8}|\d{4}-\{7,8}', re.I)
content = 'Hello, my phone number is 0371-86532100 .'
print(re_phone.findall(content))
#['371-86532100']

match()

2.match()
re.m atch ˯Ťķ{s|ŉĿĝç˦ú\¾V²®¯nŉĿĝç˦úK
Ȍ²m atch ()m¬-none
m atch &V®âE
re.match(pattern, string, flags=0)
p atte rnzh˦úĊ³zȧV²strin gzhp˦ú{s|
flagsĻȫĝ²6ü¶<Ċ³zȧV˦úOV²®Eťêeë±²59˦ú
ĩ

print(re.match('python',"Python,I like!",re.I)) #<re.Match object; span=(0, 6),
match='Python'>
print(re.match('Math',"Python,I like!")) #None

re¾ ^£˫įɘM atch*+ňO&²6üƮħ˦ú·¯^Ñȵª

search()

search()
m atch Ħ˦úZ Ŀ{s|n²se arch-t˦úZĪïĝç
{s|ĦpȜĹ\˦ú³¬-\˦ú*+²ť³¬-fN one
se arch &V®âE
re. search(pattern, string, flags=0)

print(re.search('python',"Python,I love python!",re.I)) #<re.Match object;
span=(0, 6), match='Python'>
print(re.search('Math',"Python,I love python!",re.I)) #None
}ïE
m atchYse arch˰˦úK²m\m atch o b je ct*+²¦*+ijsp anY
m atch\ĕe²Ì^sp anzh˦ú*+ÆƂĻZ^¢ĝçāȠfb0²
6c²m atchzh˦ú*+Ũ¤¥²āȠf'Python'

findall()

4.findall()
m atchĦ˦úZ Ŀ²se archÔĦ¬-\˦úȵ²®¯*Z
op>9ȗZ˦ú²³opM6fin d allYfin d ite rre¾ ^fin d all6
üÆ{s|^ȜĹƵąĊ³zȧV˦úŦ|
fin d all &V®âE
re.findall(pattern,string,flags=0)
höE
print(re.findall(r'Python', 'Python! I love python.', re.I)) #['Python', 'python']

finditer()

5. finditer()
re¾ ^fin d ite rfin d allƔȗň²êǩÆüfin d alltĬ
zµV¬-²ǧfin d ite rtŰØµV¬-·¯
&V®âE
re.finditer(pattern,string,flags=0)
höE
print(re.finditer(r'Python', 'Python! I love python.', re.I)) #<callable_iterator
object at 0x000001A33E8AA848>

split()

6. split()
splitÙd{s|ȻɥĊ³zȧV˦úŦ|>9eº
&V®âE
re.split(pattern,string,maxsplit=0,flags=0)
p atte rn,strin gYflagsjm atchjň
m axsp litŀëeʐƄ²Ľľf0²zhnƨ<eʐƄ
höE
print(re.split(r'\d+','one1two2three3four4five5')) #['one', 'two', 'three', 'four',
'five', '']

sub()

7. sub()
sub6üǵč{s|^Ċ³zȧV˦úŦ|
&V®âE
re.sub(pattern,repl,string,count=0,flags=0)
Ì^p atte rnĊ³zȧV²replpǵčK{s|²strin gpĈǵč{s
|²
co u n tŀëǵčƄ²Ľľf0²zhnƨ<ǵčƄ
höE
print(re.sub(r'\d+', '_', 'one1two2three3four4five5'))
#one_two_three_four_five_

subn()

8. su b n () su b nsu bƔȗň²Ħsu b nÙtŔåµV¬-ǵč¿
Ą{s|YǵčƄ &V®âE
re.subn(pattern,repl,string,count=0,flags=0) jsu bň höE
print(re.subn(r'\d+', '_', 'one1two2three3four4five5'))
#('one_two_three_four_five_', 5) print(re.subn(r'', '', '
ƥȿ˱²˲ʿ
')) # ('ƥȿ˱²˲ʿ', 4)

6.4 ±· 176

6.5 L¿ÀÁ 176

L¿ğƖ

,˳˴˵ó
ƞƟoƛE
˴˵óĔžČą˴˵ɽȁ˶)bSʚźɽ˷˶)cw˸˹˶)wn˺˻˼˽
óS
nZÊ6 ²˾˿ẁ́ˀǽŮĔÙǰɘ˴˵ó²t̂nūI̃̄˾
˿ẁ́CD
Ÿ˾˿ẁ́ǰçį˴˵ó²6Į ¤¥˓ją˴˵ó²˾˿Ỳ́ÙdÌ
ůf
nZÊ6 ²̅Ə¤¥Ĭ²SM6“*”ǵčÌ^˴˵ó
â(dM6P yth o n]ğ,˳˴˵ó
èƫşeƬE
ğ,˳˴˵óǑǒEôÈǰ˴˵ó4²¿*6Į ¡ ;>9
ȱȲ²
®¯Ì^ij˴˵ó4^ó̇²Ć,replace()O&M6“*”ǵč˴˵ó

se n sitive = ['ʚɽ˷','˸˹','n˺˻ó̇','nZÊ']
co n te n t= in p u t('Ɔ ¡ōȌE')
for c in sensitive:
if c in content:
result=content.replace(c,'*')
content=result
print(content)
# Ɔ ¡ōȌEʚɽ˷nĊƽ
# *nĊƽ

7 I/O ŧć 180

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7.1.1 ȲȆȹ#$ 180

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7.1.1 3 ůÞžǒɸťf̈*ǒɸ 185

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7.1.1 6 ƮħZ[R 186

7.1.1 7 ƮħZ[ǒɸ 186

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7.2.2 w ith ; 188

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7.3.2 ďɶª 192

7.3.3 M6C SVVɝĳɶwďɶª 192

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7.4.1 ŧć 194

7.4.2 ŧće3 194

7.5 ŧćǼǽ 195

7.5.1 try except 196

Æ] I^²ȱţǼǽŧć²-tĆ,try-e xce p t ;yğ²Īī¢Ætry
;Ł¤ŧćõ-tĈȱţĹ
&E
try:
< ;> #?9ǩ_
e xce p t <ŧć3R{> [as R]E
< ;> #®¯ÆtryĕŬuIįŧć²ź9¦ ;
else:
< ;> #®¯ŠąŧćIʆ
finally:
< ;> # ɧ˾ťąŧćIʆõÙź9_

®¯p*5»ŧć>9ȹǼǽ²ɛ6®â &VE
try:
[;]
e xce p t (Exce p tio n 1²Exce p tio n 2²Exce p tio n 3²…):
¢ŧć(e xce p tio n)¿Ǽǽ_
5\ŧć3R6l“²”º ²ŧć3È¿Ķ]nÙą̃̄²ʢȓȻɥŀ
È¢ŧć*+>9˦ú
Try-e xce p t ;ũą\Ŧ;e lseŦ;Yfin allyŦ;ElseŦ;®¯try¿ ;
^ŠąIʆŧć²³ź9e lse ;¤¥Fin llayŦ;ɧ˾ťIʆŧć²õ
ÙĈź9}ïEe lseŦ;ǅĢģąe xce p tŦ;Fin allyŦ;«npƛǅ(Ģģą
e xce p tŦ;Ye lseŦ;

7.5.2 else 196

7.5.3 finally 197

7.5.4 raise 198

7.5.5 Þ 198

7.5.6 Ûŧć 199

7.6 ±· 199

7.7 L¿ÀÁ 199

ğƖ

# 1cÛ\ŧć3O u tO fStack
# 2cƆǶ×ğöbuyū̉̊̋̌ÄƧ¢4ýÖÙ̍¢O u tO fStackŧć
# 3cǪ̎&Æƕ]^\_Ùuŉ]̍¢O u tO fStack
# 4c̏Ʈ¦ŧć²« ¢ŧćȪ¸¹
class CandleShop:
name = "Here's a Hot Tip: Buy Drip Candles"
def __init__(self, stock):
self.stock = stock
def buy(self, color,num):
self.stock[color] = self.stock[color] - int(num)

class OutOfStack(Exception):
pass
buy_bluenum=7
candle_shop = CandleShop({'blue': 6, 'red': 2, 'green': 0})
candle_shop.buy('blue',buy_bluenum)
stock=candle_shop.stock
try:
for value in stock.values():
if value<0:
raise O utO fStack('ʸʹÄƧ¢4ý²ɧ&̉̊ų')
except OutOfStack as e:
print(e)

2.ą\u se r.csvZ[²¤¥®âE
name,username,email
̐̑,yangy,yangy@ sina.com
̒Ŧ̓,jiazh,jiazh@ 126.com
ü̔,yuf,yuf@ 163.com
̖̗̕,tianych,tianych@ sina.com
1)t{ĭVħcsvZ[«ʙ̘¢ò\ƘR{YǃŦ̙[șȚ
2)ĄTu se rco p y.csvZ[²du se r.csvZ[¤¥ȻɥcsvZ[¡OV¡>
u se rco p y.csv^

import csv

with open('user.csv', mode='r', encoding='utf-8') as f:
re = csv.DictReader(f)
for row in re:
print(row)
with open('user.csv', mode='r', encoding='utf-8') as f1:
with open('usercopy.csv', mode='w', encoding='utf-8') as f2:
for line in f1:
f2.write(line)

8 5ä5> 202

8.1 ä> 202

8.1.1 ä 202

8.1.2 > 202

8.2 5ä 203

8.2.1 5äÐTY̚Ü 203

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8.2.3 ̛ɗä 207

8.2.4 5äņ 208

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8.3.3 5>ņ 218

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1 0.4 ±· 275

ª-ɨÓ

ˈ̣Ǡ

import numpy as np
import matplotlib.pyplot as plt
# 6yĊćgh^Zb̤c ć6{ikE Kaiti-̥ġ FangSong-̧̦ġ
M icro so ft Y aH e i-̨̩̪̤
plt.rcParams['font.sans-serif'] = 'SimHei'
6yĊćghċl
plt.rcParams['axes.unicode_minus'] = False

x=np.arange(0,12,0.2)
y=np.cos(x)
plt.plot(x,y,'bs--')

p lt.title ("ˈ̣̫äǠ")
p lt.xlab e l("x̬")
p lt.ylab e l("y̬")

plt.savefig("picture1",dpi=300)
plt.show()

̭̮Ǡ

import numpy as np
import matplotlib.pyplot as plt
plt.rcParams['font.sans-serif'] = 'SimHei'
plt.rcParams['axes.unicode_minus'] = False

bins=[0,60,70,80,90,100]
hist=[2,14,27,17,3]

bins[0]=50
x=bins[0:-1]
y=hist
plt.axis([40,100,0,40])
p lt.xticks(x,['6 0etâ','60-70e','70-80e','80-90e','90-100e'])
plt.bar(x,y,width=5,color='steelblue',align='center')

for x,y in zip(x,hist):
plt.text(x,y+0.05,str(y),ha='center',va='bottom')

p lt.title ("KȺeĤǠ")
p lt.xlab e l("eō")
p lt.ylab e l("Ƙ")
plt.show()

import numpy as np
import matplotlib.pyplot as plt
p lt.rcP aram s['fo n t.san s-se rif'] = 'Sim H e i' # MǠµ^^ZĊć _gh
p lt.rcP aram s['axe s.u n ico d e _m in u s'] = False # M̯Ļ̬̰AẕĊćghĊċl
# ķcsvZ[^ħ ʆKȺ
arrstd=np.loadtxt("ScoreTest.csv",dtype=bytes,delimiter=',',skiprows=1).astype
(str)

# 2. n d arrayªŃń
sarr = arrstd [:, 7:8] #ƮħƵą ʆƐKȺĬªsarr
tarr = arrstd [:, 1:7] #ƮħƵą ʆÑ\ÁƂȇeª
sarr = sarr.astype(float)
tarr = tarr.astype(float)
hist,bins=np.histogram(sarr, bins=[0,60,70,80,90,100],range=(0,100))
# ęò̲ÁƂ#ŋe
tmean = tarr.mean(axis=0)
# ęȇeʁE#ŋe/ƐeÅ
ts= n p.array([2 0,1 0,1 0,2 0,2 0,2 0]) #ÁƂeÅåE
tdfl = tmean/ts
b in s[0]= 5 0 # 5 0z60etâKȺ²\h iståň
x = bins[0:-1]
y = hist
# Ƕ×̯Ļ̬̰AÅ
# axisr§\list²ǰ̴̳̯Ļ̵A²listÑ\eǩz[XĿ̰A²X
ƎƏ̰A²YŉĿ̰A²YƎƏ̰A]
p lt.axis([4 0,1 0 0,0,1 0]) #ǰç̯ĻÅŁ, x̬Ł[4 0,1 0 0],y̬Ł[0,1 0]
p lt.xticks(x,['6 0etâ','60-70e','70-80e','80-90e','90-100e'])
plt.bar(x, y, width=5, color='steelblue', align='center')
# ÆĪïĝçZ
# p lt.te xt(x̳̯Ļ²y̴̯Ļ²ghª²̶#*ŏ²̷Í*ŏc
fo r x,y in zip (x,h ist): # d-Ű*+ʙiK\Ŕå
plt.text(x, y+0.05, str(y), ha='center', va= 'bottom')
# ɑ̸̹̺̀
plt.grid(True)
# ǰçĻÁË̯Ļ̬Rĉ
p lt.title ("KȺeĤǠ")
p lt.xlab e l("eō")
p lt.ylab e l("Ƙ")
# ghǠz
plt.show()

̻ȧǠ

import matplotlib.pyplot as plt
import numpy as np
plt.rcParams['font.family'] = ['SimHei']
plt.rcParams['axes.unicode_minus'] = False

#ķcsvZ[^ħ ʆKȺ
arrstd=np.loadtxt("ScoreTest.csv",dtype=bytes,delimiter=',',skiprows=1).astype
(str)

# 2. n d arrayªŃń
sarr = arrstd [:, 7:8] #ƮħƵą ʆƐKȺĬªsarr
tarr = arrstd [:, 1:7] #ƮħƵą ʆÑ\ÁƂȇeª
sarr = sarr.astype(float)
tarr = tarr.astype(float)
hist,bins=np.histogram(sarr, bins=[0,60,70,80,90,100],range=(0,100))
# ęò̲ÁƂ#ŋe
tmean = tarr.mean(axis=0)
# ęȇeʁE#ŋe/ƐeÅ
ts= n p.array([2 0,1 0,1 0,2 0,2 0,2 0]) #ÁƂeÅåE
tdfl = tmean/ts
b in s[0]= 5 0 # 5 0z60etâKȺ²\h iståň
x = bins[0:-1]
y = hist
#ŷTª
angles= np.linspace(0,2*np.pi,6,endpoint=False)
data= tdfl
labels = ['ŌˁÁ', '˚Á', 'ůÞÁ', '̼Á', '] Á', ']˚Á']
#̽<̯̾ĻǠ
p lt.p o lar(an gle s,d ata, "b o -",lw = 1) #̯̾Ļ̽Ǡ² bo-zhV{s|
#ǰçĻÁË̯Ļ̬Rĉ
p lt.title ("ȇeʁ̻ȧǠ")
p lt.th e tagrid s(an gle s * 1 8 0 / n p.p i, lab e ls, y= 0.0 2) # ǰç̾ŜĻ̱
#ȨĒªqäEM6co n cate n ate ()qr<ĳȨĒª
an gle s= n p.co n cate n ate ((an gle s,[an gle s[0]])) #ŜAªôŊňr²̫äȨĒ
d ata= n p.co n cate n ate ((d ata,[d ata[0]])) #ȇeʁªôŊňr²̫äȨĒ
#˚˛ȨĒqäêȕ
plt.fill(angles,data, "r",alpha=0.3)
#Ƕ×ĻÁĝç
p lt.title ("ȇeʁ̻ȧǠ",y=1.06,fontsize=15) #ǥ°ĻÁghĝçËZ{ë±
#Ƕ×̯̾ĻŁ
p lt.ylim (0,1) #ǰç̯̾Ļ̬Łf0Ĺ1
#ɑªªgh
for x,y in zip(x,hist):
plt.text(x, y+0.09, round(y,2), ha='center', va= 'bottom')
plt.show()